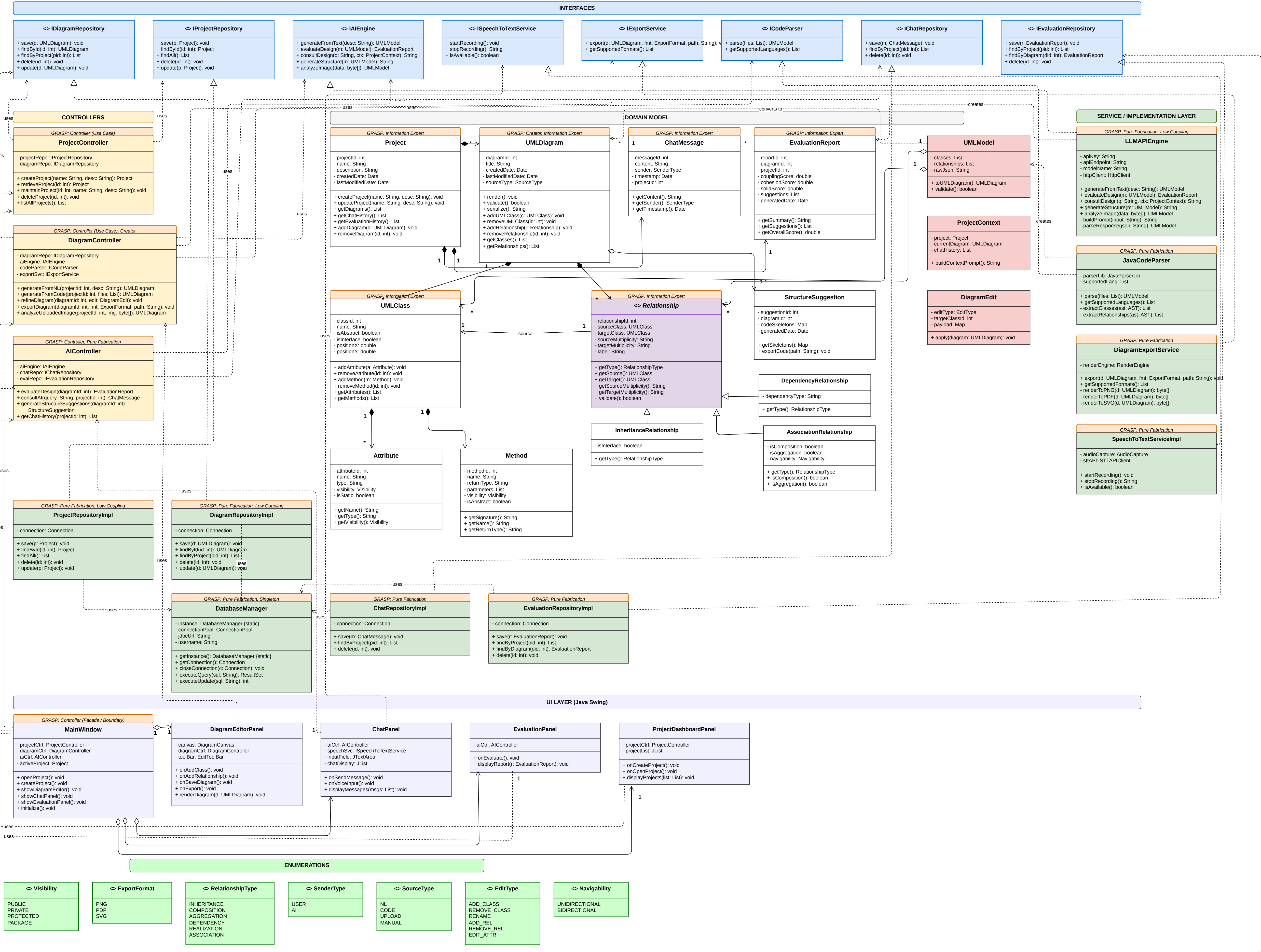
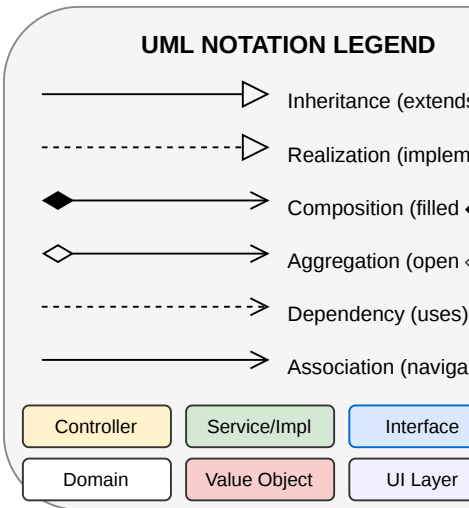




GRASP Patterns Summary

Controller: ProjectController, DiagramController, AIController, MainWindow
Handle system events; delegate work to domain objects.

Information Expert: Project, UMLDiagram, UMLClass, Attribute, Method, Relationship, ChatMessage, EvaluationReport
Own and manage their information.

Creator: DiagramController creates UMLDiagram, UMLDiagram (Composition)
owns UMLClass + Relationship, UMLClass (Composition) owns Attribute + Method.

Pure Fabrication: All Repository Impl, LLMAPEngine, JavaCodeParser, DiagramExportService, SpeechToTextServiceImpl, DatabaseManager
Abstract classes supporting low coupling and high cohesion.

Low Coupling: Controllers depend only on interfaces, never on concrete.

High Cohesion: Each class has exactly one well-defined responsibility.

Polymorphism: 7 interfaces with interchangeable implementations, abstract Relationship with 3 concrete subtypes.